

for semiconductor production, has developed technology to separate chemicals from wastewater, enabling solvent recycling and recovering valuable materials, such as gold. As manufacturers develop more minor, more advanced chips, this approach could lower emissions per microchip—an objective for Infineon, which is expanding its production of energy-efficient silicon carbide and gallium nitride chips while planning new factories.

Renewable energy integration is another key strategy for reducing carbon footprints. While dependence on renewables varies by country, many facilities have installed solar panels and other energy systems, achieving high renewable energy usage rates. Despite these advances, the industry remains energy-intensive, underscoring the need for ongoing efficiency improvements.

TSMC has implemented energy-saving measures, including standby and off modes for equipment, reducing consumption when machines are idle. However, fluorinated gases—used in etching processes—remain a significant source of emissions due to their long atmospheric lifetimes and high global warming potentials. Continued innovation in water and energy efficiency is essential to mitigating the environmental impact of semiconductor manufacturing.

Green tech's tightrope

Cost remains a significant hurdle in sustainable manufacturing, particularly for printed circuit boards (PCBs). Adopting new processes or materials requires substantial capital investment, creating barriers for traditional companies. However, newer firms can more easily embrace sustainable practices as they are unburdened by legacy systems. This flexibility enables them to implement innovative techniques, such

Cutting e-waste with smart manufacturing

- **Early fault detection.** Integrating early testing in manufacturing minimises e-waste by catching defects upfront, preventing resources from being wasted on faulty products. This conserves materials and reduces the environmental impact of discarded electronics.
- **Managing excess inventory.** Excess stock leads to significant waste, with about 1% of electronics revenue lost to landfill-bound components. To tackle this, distributors are refining stock management and purchasing surplus from smaller manufacturers, creating new revenue streams while reducing waste.
- **Reusing manufacturing equipment.** Refurbishing and reselling manufacturing equipment extends its lifespan and lowers environmental impact. Companies like Nikon reclaim and refurbish used equipment, reducing the need for new production. The shipment of ASML's EUV machines alone requires three Boeing 747s, illustrating the substantial footprint of new equipment manufacturing.
- **Recycling and recovery at end-of-life.** End-of-life electronics present valuable recycling opportunities. Recovering components and precious metals for reintegration into manufacturing reduces costs and reliance on raw material extraction. This approach conserves resources and curtails the environmental toll of mining, supporting industry sustainability.

as additive manufacturing, which produces far less waste than traditional etching.

Scaling these sustainable methods comes with challenges. Technology validation requires rigorous testing and partnerships to establish effectiveness and trust. Achieving economies of scale is another obstacle, as sustainable materials often incur higher costs initially due to the absence of large-scale production efficiencies. However, if these materials provide long-term benefits—such as the recyclability of precious metals—the higher initial investment may be justified.

Furthermore, integrating these new methods into existing systems also poses difficulties. Materials that successfully transition typically maintain compatibility with traditional manufacturing techniques. For example, polylactic acid—a sustainable substrate—endures temperatures up to 140°C, making it viable for alternative methods such as laser sintering while easing adoption into existing production lines.

Germany exemplifies progress in waste management and electronics recycling. The country's robust infrastructure for separate waste collection and processing

sets a global standard, supporting a circular economy that extracts and reuses valuable materials at scale. As companies worldwide grapple with the costs and complexities of sustainable adoption, success will likely depend on innovative materials, strong legislative frameworks, and strategic partnerships.

The environmental footprint of PCB and IC manufacturing is undeniable, given its resource-intensive nature. TSMC's high energy and water consumption underscores the urgency of sustainable practices in an industry integral to modern technology. While challenges such as waste, resource depletion, and pollution persist, the industry's shift towards sustainable processes—driven by advanced materials and recycling innovations—signals progress towards a more responsible future. Continued collaboration and stringent environmental standards will ensure that emerging technologies align with manufacturing needs and broader sustainability goals, ultimately reducing the tech industry's ecological impact. **EFY**

Co-author, Nidhi Agarwal, is Senior Technology Journalist at EFY with a deep interest in embedded systems, development boards, and IoT cloud solutions.